Three Methods for Solving Systems of Linear Equations

Graphing • Substitution • Elimination

A system of linear equations asks for an ordered pair (x, y) that makes both equations true. Graphing, substitution, and elimination are three ways of finding the same solution. The best method depends on what the equations look like and what we need to know.

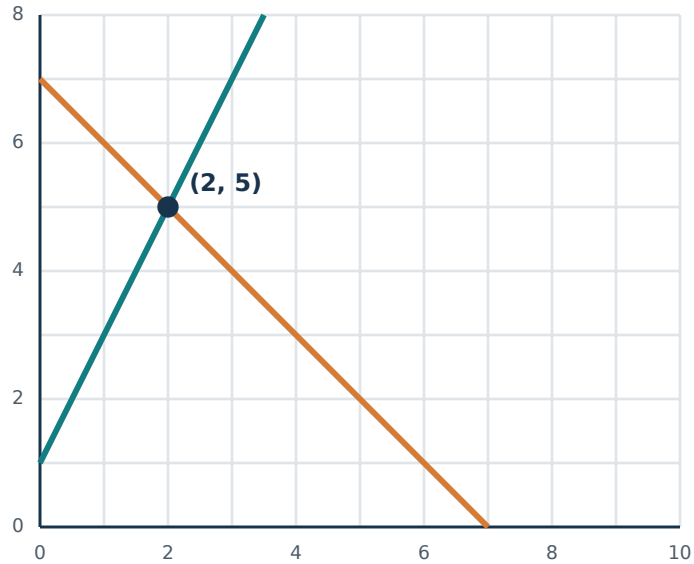
One system, three methods: y = 2x + 1 and y = −x + 7. The solution is (2, 5), because 5 = 2(2) + 1 and 5 = −2 + 7.

1 Graphing — see where the lines meet

Why it works: Every point on a line satisfies its equation. The intersection lies on both lines, so it satisfies both equations.

Steps: Write each equation in a graphable form; plot both lines on the same axes; locate their intersection; check that point in both equations.

For our example, graph y = 2x + 1 (slope 2, y-intercept 1) and y = −x + 7 (slope −1, y-intercept 7). The lines meet at (2, 5).



Graphing is best when: the equations are easy to graph, the intersection is at clear grid coordinates, or the goal is to understand what a solution means. It is especially useful for seeing one solution (crossing lines), no solution (parallel lines), and infinitely many solutions (the same line).

Limitation: A graph can only give an estimate when the intersection has fractional or non-integer coordinates. Use algebra when an exact answer matters.

2 Substitution — replace equal quantities

Why it works: If y equals an expression, that expression can replace y anywhere else. Replacing one equal quantity with another does not change the truth of an equation.

Worked example: one variable is already isolated

$$y = 2x + 1$$

$$y = -x + 7$$

Both expressions equal y. At the intersection they must equal each other. This is substitution in its most convenient form.

$$2x + 1 = -x + 7$$

$$3x + 1 = 7$$

$$3x = 6$$

$$x = 2$$

Put x = 2 into either original equation:

$$y = 2(2) + 1 = 5$$

Solution: (2, 5)

Check in the other equation: $5 = -2 + 7$, which is true.

Another example: substitute into an equation

$$y = x + 3$$

$$2x + y = 12$$

Replace y in the second equation with x + 3:

$$2x + (x + 3) = 12$$

$$3x + 3 = 12$$

$$3x = 9$$

$$x = 3$$

$$y = 3 + 3 = 6$$

Solution: (3, 6). Check: $2(3) + 6 = 12$ and $6 = 3 + 3$.

Substitution is best when: a variable is already alone (such as y = 2x + 1), a coefficient is 1 or −1 so isolating a variable is easy, or both equations already equal the same variable. Setting the two y-expressions equal is a particularly efficient form of substitution.

Watch out: Use parentheses when replacing a variable with a multi-term expression. For example, x − y = 4 and y = 2x + 1 becomes x − (2x + 1) = 4, not x − 2x + 1 = 4.

3 Elimination — combine equations to remove a variable

Why it works: We may add or subtract the left sides of two true equations and do the same to their right sides. If a variable has opposite coefficients, it disappears in the sum. Multiplying an entire equation by a nonzero number preserves its solutions.

Worked example: coefficients already cancel

$$x + y = 8$$
$$x - y = 2$$

Add the equations vertically. The +y and −y terms cancel:

$$2x + 0y = 10$$
$$2x = 10$$
$$x = 5$$

Substitute x = 5 into x + y = 8: y = 3. Solution: (5, 3). Check: 5 + 3 = 8 and 5 − 3 = 2.

Worked example: multiply before eliminating

$$2x + y = 7$$
$$3x - 2y = 4$$

Multiply the entire first equation by 2 to create opposite y-coefficients:

$$4x + 2y = 14$$
$$3x - 2y = 4$$
$$\text{———————}$$
$$7x = 18$$
$$x = 18/7$$

Now use 2x + y = 7:

$$y = 7 - 2(18/7) = 13/7$$

Solution: (18/7, 13/7)

Check: 2(18/7) + 13/7 = 49/7 = 7; 3(18/7) − 2(13/7) = 28/7 = 4.

Elimination is best when: the equations are in standard form (Ax + By = C), coefficients already match or are opposites, or small whole-number multipliers will make them cancel. It often avoids awkward fractions until the end.

Watch out: Multiply every term, including the constant. When subtracting equations, subtract the entire second side; parentheses help prevent sign mistakes.

How do I choose the best method?

What you notice	Try first	Why
Easy to graph; visual meaning matters	Graphing	Shows the shared point and number of solutions
One variable is already isolated	Substitution	Replace directly; little rearranging
Both equations equal y	Substitution	Set the right-hand expressions equal
Standard form; coefficients cancel	Elimination	Add or subtract to remove a variable
Standard form; small multipliers make a match	Elimination	Scale equations, then cancel
Intersection is not on a grid point	Algebra	Substitution or elimination gives an exact answer

Practice: choose a method before solving

1. $y = 3x - 2$ and $y = -x + 10$
Suggested choice: Substitution: both equal y; set expressions equal.

2. $2x + 3y = 12$ and $2x - y = 4$
Suggested choice: Elimination: subtract to remove x.

3. $y = x + 1$ and $y = -x + 5$
Suggested choice: Graphing: lines are easy to draw; intersection is (2, 3).

4. $4x + 3y = 17$ and $2x - 3y = 1$
Suggested choice: Elimination: y-coefficients are opposites.

Questions to ask the student

What does the solution represent? Which method requires the least algebra here? Can you explain why the operation preserves the solution? How would you check the answer? Would a different method also work?

Teaching takeaway: There is no single “correct” method for every system. Choose the method that makes the structure of the equations easiest to use, then verify the ordered pair in both original equations.